

MACHINE LEARNING

IMAGE RECOGNITION

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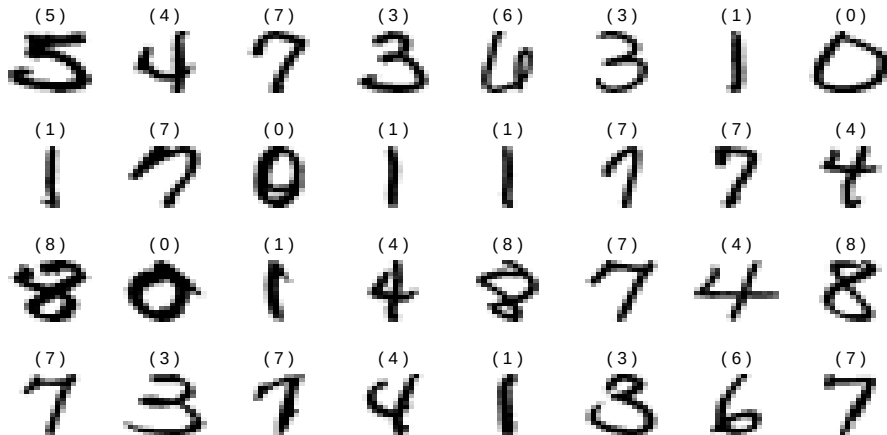
MASTER IN BUSINESS ANALYTICS



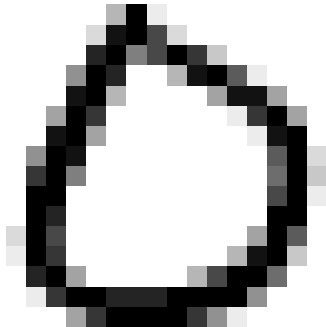
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Example: handwritten digit classification

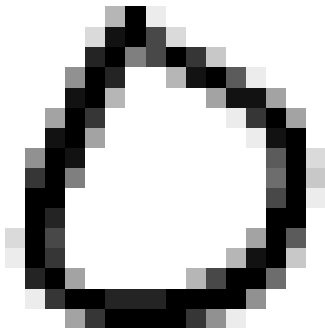
Dataset from [\[ESL\]](#) webpage. Normalized handwritten digits (16×16 grayscale images), automatically scanned from envelopes by the [U.S. Postal Service](#). Observations consists of the digit $y_i \in \{0, \dots, 9\}$ and the $p = 256$ grayscale values $\mathbf{x}_i = (x_{i1}, \dots, x_{i256})$. The training and test sets contain 7290 and 2006 examples, respectively.



Digital black-white images

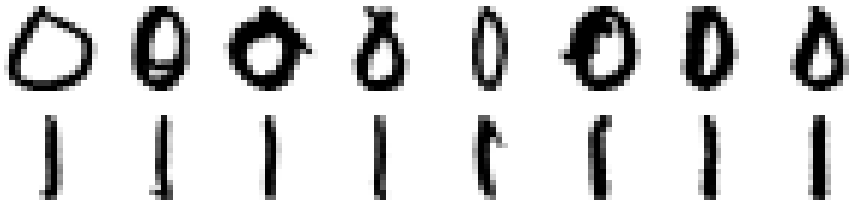


Digital black-white images

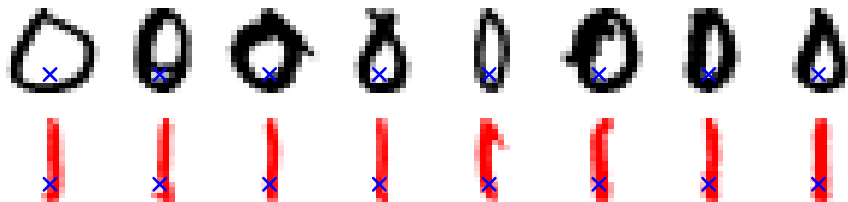


0	0	0	0	0	70	240	33	0	0	0	0	0	0	0	0	0	0
0	0	0	0	36	230	255	181	43	0	0	0	0	0	0	0	0	0
0	0	0	3	218	255	131	180	248	199	59	0	0	0	0	0	0	0
0	0	0	112	255	212	12	4	82	230	253	169	26	0	0	0	0	0
0	0	8	225	255	81	0	0	0	11	95	234	227	99	1	0	0	0
0	0	95	255	196	2	0	0	0	0	0	21	198	255	91	0	0	0
0	8	235	255	94	0	0	0	0	0	0	0	22	235	244	7	0	0
0	109	255	225	3	0	0	0	0	0	0	0	0	0	157	255	49	0
0	200	255	122	0	0	0	0	0	0	0	0	0	0	150	255	55	0
10	247	252	17	0	0	0	0	0	0	0	0	0	0	173	255	28	0
17	253	211	1	0	0	0	0	0	0	0	0	0	8	239	239	5	0
51	255	180	0	0	0	0	0	0	0	0	0	0	92	255	162	0	0
28	254	204	0	0	0	0	0	0	0	0	4	72	230	253	60	0	0
0	208	244	88	15	0	0	0	13	85	176	255	253	152	0	0	0	0
0	26	194	255	252	206	206	206	240	255	255	241	116	0	0	0	0	0
0	0	6	91	190	255	255	255	255	192	105	24	0	0	0	0	0	0

Digital black-white images

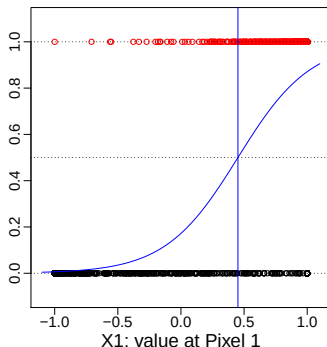


Two-class classification: one predictor

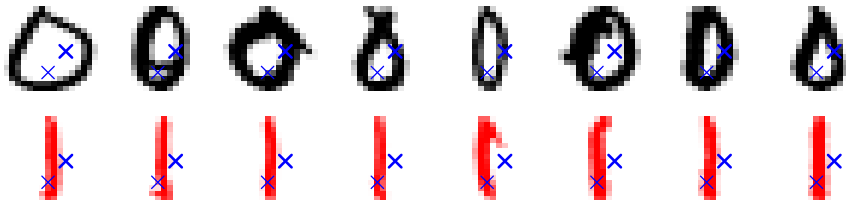


- ▶ Value at Pixel 1 (blue cross) is the predictor
 $X_1 = \text{"value at Pixel 1"} \in [-1, 1]$
- ▶ The class is $Y \in \{0, 1\} = \{\text{zero}, \text{one}\}$
- ▶ We apply logistic regression to obtain a **classifier** based only on Pixel 1:

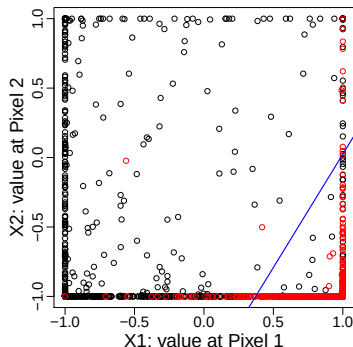
$$\hat{G}(x) = \begin{cases} \text{zero} & \text{if (Pixel 1)} < 0.45 \\ \text{one} & \text{otherwise.} \end{cases}$$



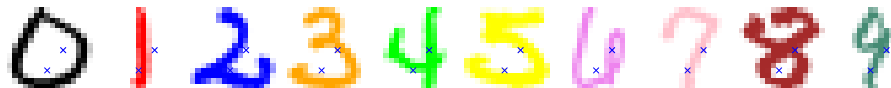
Two-class classification: two predictors



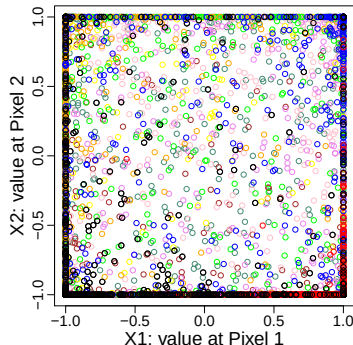
- ▶ $p = 2$: Predictors are
 $X_1 = \text{"value at Pixel 1"} \in [-1, 1]$
 $X_2 = \text{"value at Pixel 2"} \in [-1, 1]$
- ▶ The class is $Y \in \{0, 1\} = \{\text{zero}, \text{one}\}$
- ▶ We apply logistic regression to obtain a **classifier** based only on X_1 and X_2 .



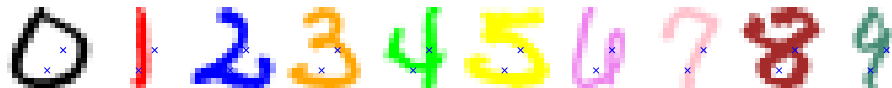
10-class classification: two predictors



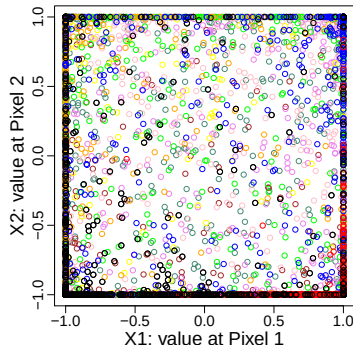
- ▶ $p = 2$: Predictors are
 $X_1 = \text{"value at Pixel 1"} \in [-1, 1]$
 $X_2 = \text{"value at Pixel 2"} \in [-1, 1]$
- ▶ The class is
 $Y \in \{0, 1, \dots, 9\} = \{\text{zero}, \text{one}, \dots, \text{nine}\}$
- ▶ We can apply **multinomial regression**, based on X_1 and X_2 , but with only two pixels this a very hard task!



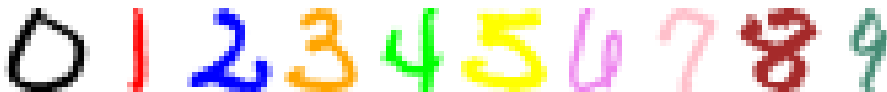
10-class classification: two predictors



- ▶ $p = 2$: Predictors are
 $X_1 = \text{"value at Pixel 1"} \in [-1, 1]$
 $X_2 = \text{"value at Pixel 2"} \in [-1, 1]$
- ▶ The class is
 $Y \in \{0, 1, \dots, 9\} = \{\text{zero}, \text{one}, \dots, \text{nine}\}$
- ▶ We can apply **multinomial regression**, based on X_1 and X_2 , but with only two pixels this is a very hard task!
- ▶ Luckily we have $p = 256$ predictors (pixel values) to perform this task...



Digit classification

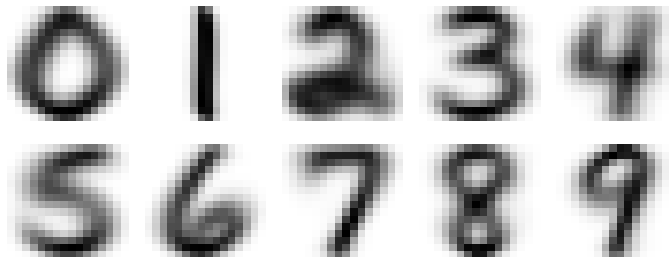


- ▶ **Training data:** $(x_1, y_1), \dots, (x_n, y_n)$, $n = 7290$, are used for model fitting/selection, where
 - ▶ predictors $x_i = \text{i-th image} \in \mathbb{R}^p$ are the grayscale values of the $p = 256$ pixels;
 - ▶ outcomes $y_i = \text{i-th digit} \in \{\text{zero}, \text{one}, \dots, \text{nine}\}$.
- ▶ **Test data:** $(\tilde{x}_1, \tilde{y}_1), \dots, (\tilde{x}_m, \tilde{y}_m)$, $m = 2006$, are not used for model fitting/selection, but only to evaluate the performance afterwards.
- ▶ linreg: direct linear regression
- ▶ LDA1: LDA based on the values x_i
- ▶ LDA2: LDA based on x_i and x_i^2
- ▶ QDA: different cov. matrix for each class
- ▶ log: logistic regression (multinomial)
- ▶ QDA is worst of the four methods since its variance is too high (overfitting).

	Training error	Test error
linreg	7.6%	13.1%
LDA1	6.2%	11.5%
LDA2	3.9%	10.2%
QDA	1.8%	13.5%
log	0.01%	11.1%

Digit classification: visualization of LDA1

- ▶ Figure shows estimates $\hat{\mu}_1, \dots, \hat{\mu}_q$ for LDA1.
- ▶ This gives an impression of the centers of the classes, i.e., how American people write in ‘average’ the 10 digits.

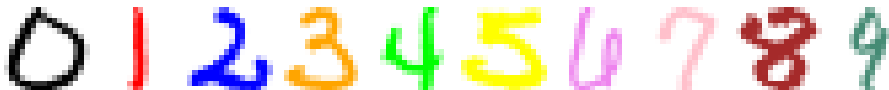


Example: digit classification

The table shows some misclassified test data from LDA1. The 10 columns on the right give the **predicted class probabilities** (in %) for $k \in \{0, \dots, 9\}$.

Obs		0	1	2	3	4	5	6	7	8	9
	(6)	59	0	0	0	0	0	40	0	0	0
	(2)	94	0	0	6	0	0	0	0	0	0
	(3)	0	0	0	11	0	0	0	0	89	0
	(3)	0	0	0	1	0	99	0	0	0	0
	(8)	1	0	0	81	1	11	0	0	5	0
	(4)	0	0	0	0	15	0	0	0	0	85
	(9)	0	0	0	0	0	0	0	100	0	0
	(2)	100	0	0	0	0	0	0	0	0	0

Digit classification with regularization



- ▶ We can update our results on the **digit classification**.
 - ▶ When estimating the logistic multinomial regression, we introduce a **ridge penalty** and a **lasso penalty** in the likelihood.
 - ▶ The value of $\hat{\lambda}$ for the tuning parameters were found by 5-fold CV.
-
- ▶ linreg: direct linear regression;
 - ▶ LDA1: LDA based on the values x_i ;
 - ▶ LDA2: LDA based on x_i and x_i^2 ;
 - ▶ QDA: different cov. matrix for each class;
 - ▶ log: logistic regression (multinomial).
 - ▶ log-ridge: logistic regression (multinomial) with ridge penalty.
 - ▶ log-lasso: logistic regression (multinomial) with lasso penalty.

	Training error	Test error
linreg	7.6%	13.1%
LDA1	6.2%	11.5%
LDA2	3.9%	10.2%
QDA	1.8%	13.5%
log	0.01%	11.1%
log-ridge	4.1%	9.0%
log-lasso	2.7%	8.8%